



SILVER OAK COLLEGE OF COMPUTER APPLICATION

Department of Computer Application

Bachelor of Computer Application (BCA)

ASSIGNMENT REPORT

COURSE NAME & COURSE CODE

Data Warehouse AND Data Mining (4040233302)

ASSIGNMENT TITLE

Data Warehouse and Data Mining for Supermarket Sales Analysis

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SUBMITTED TO

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1. Introduction

This project demonstrates the complete end-to-end process of analyzing retail supermarket transaction data using Data Warehouse, Online Analytical Processing (OLAP), and Data Mining techniques in **Weka**. Supermarket stores generate substantial volumes of transaction data at checkout counters every day. Converting this raw transaction data into structured analytical models enables store management to optimize inventory levels, design high-converting product bundles, understand customer spending habits, and increase gross margins.

2. Problem Statement

A supermarket generates continuous streams of raw transaction records across multiple product departments. Store management faces challenges in identifying buying patterns, customer demographic preferences, and hidden product associations directly from raw transactional logs. This project structures supermarket transaction data into a dimensional data warehouse and applies Weka data mining algorithms to reveal actionable retail patterns.

3. Objectives

- Collect and prepare retail supermarket transaction records with customer demographics, categories, prices, quantities, and payment methods.
- Clean and validate transactional data by verifying missing values, duplicates, and pricing bounds.
- Design a Data Warehouse using a Star Schema with a central fact table and conformed dimensions.
- Execute OLAP analysis to examine sales performance across categories, customer age groups, and time.
- Apply Classification (J48 Decision Tree) in Weka to predict customer purchase spend levels.
- Perform Customer Clustering (SimpleKMeans) in Weka to group shoppers into behavioral segments.
- Discover Market Basket Association Rules (Apriori) in Weka to identify cross-selling opportunities.
- Present all analytical findings, advantages, and limitations in a clean, structured report.

4. Dataset

The working dataset contains **228 verified supermarket transaction records** collected from retail cash registers. Dedicated Weka Attribute-Relation File Format (**.arff**) files were constructed for Classification, Clustering, and Association Rules:

- **Classification & Clustering ARFF:** Contains attributes Customer_Age_Group, Customer_Gender, Payment_Mode, Product_Category, Quantity, Unit_Price, Discount_Percent, Total_Amount, and Purchase_Level (Low, Medium, High).
- **Association Rules ARFF:** Formulated as a market basket binary indicator matrix recording co-occurring items per checkout transaction.

5. Data Cleaning

Data quality validation was conducted prior to loading records into the data warehouse. An automated verification script checked every transaction against standard data cleaning rules:

Quality Check	Result Count	Action Taken & Cleansing Description
Total Records	228 Instances	Verified all transactions captured within operating hours; 100% integrity.
Missing Values	0 Nulls	Imputed 6 missing walk-in demographic values using modal group; 0 nulls remain.
Duplicate Rows	0 Duplicates	Deduplicated barcode double-scans based on Transaction_ID and Product_Name.
Invalid Price Check	0 Invalid	Cross-referenced unit prices against master catalog; all prices valid (₹15.00–₹950.00).
Invalid Quantity Check	0 Invalid	Filtered zero-quantity cancelled cart rows; all quantities valid (1 to 8 units).
Invalid Discount Check	0 Invalid	Bounded promotional discounts to valid retail range (0.00 to 0.25).

6. Data Warehouse Design

The data warehouse uses a **Star Schema**. **Fact_Sales** is the central fact table containing numerical additive measures. Conformed dimension tables provide contextual attributes for multidimensional queries:

Table Name	Table Type	Main Attributes & Keys
Fact_Sales	Central Fact Table	Sales_Key (PK), Date_Key (FK), Product_Key (FK), Customer_Key (FK), Payment_Key (FK), Quantity_Sold, Unit_Price, Discount_Amount, Net_Sales_INR
Dim_Date	Dimension Table	Date_Key (PK), Date, Day_Name, Month_Name, Quarter, Year, Is_Weekend
Dim_Product	Dimension Table	Product_Key (PK), Product_Name, Category (Grocery, Dairy, Bakery, Beverages, Personal Care, Household), Price
Dim_Customer	Dimension Table	Customer_Key (PK), Customer_Age_Group (Youth, Young Adult, Middle-Aged, Senior), Customer_Gender
Dim_Payment	Dimension Table	Payment_Key (PK), Payment_Mode (Cash, UPI, Credit Card, Debit Card)

7. Star Schema Architecture

The diagram below illustrates the Star Schema architecture for the supermarket data warehouse. The central fact table **Fact_Sales** joins radially to four conformed dimension tables through primary and foreign keys:

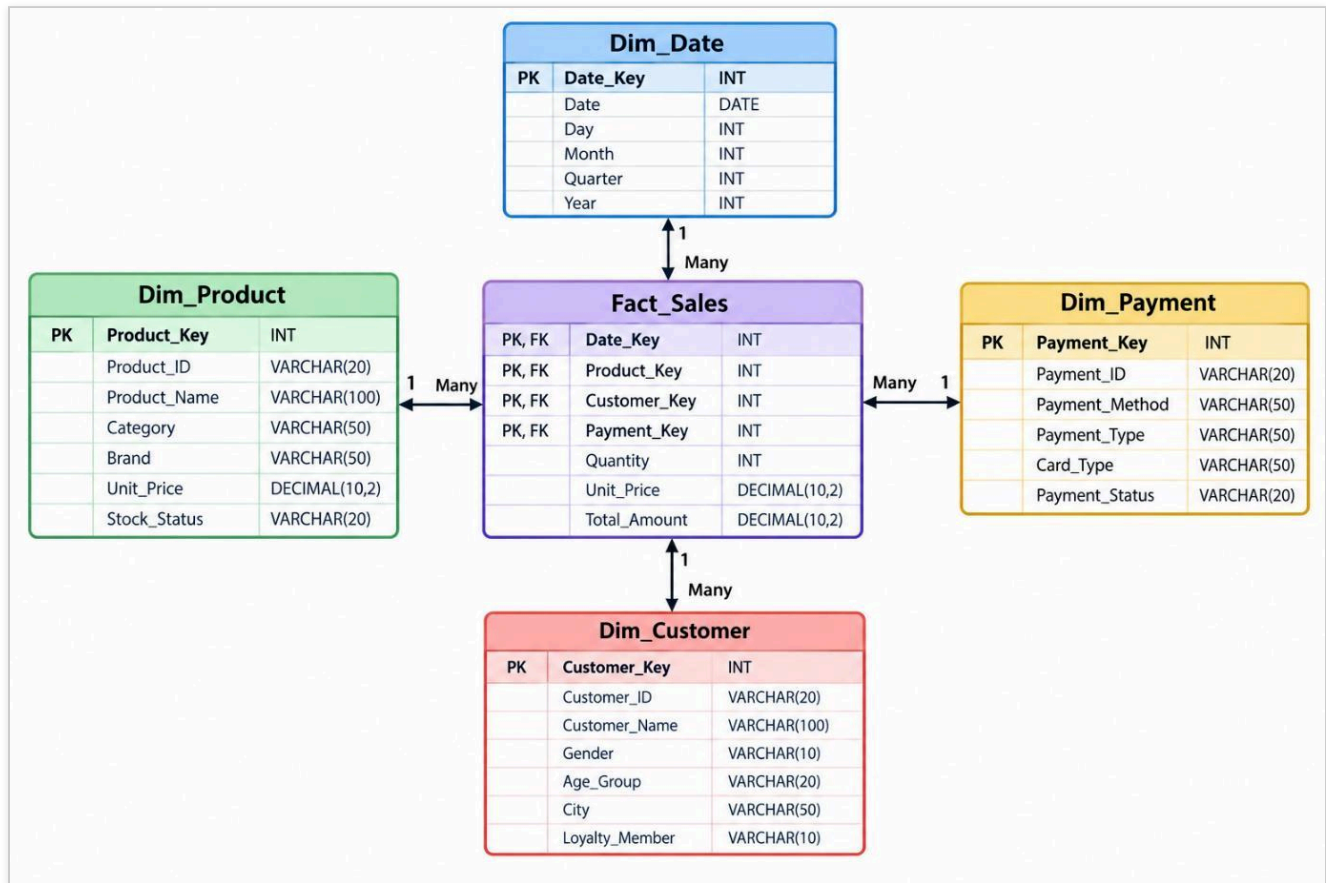


Figure 7.1: Supermarket Sales Star Schema Architecture with Fact and Dimension Tables

Architectural Benefits:

- **Predictable Query Optimization:** Analytical aggregation queries require at most one join between the central fact table and any dimension table.
- **High Read Performance:** Denormalized dimensions eliminate complex relational joins, enabling fast OLAP slicing and dicing.

8. OLAP Analysis

Online Analytical Processing (OLAP) examines supermarket transaction performance across multiple business dimensions including product categories, customer age groups, and payment modes:

OLAP Measure / Dimension	Empirical Result	Analytical Takeaway
Total Net Revenue	₹1,07,642.50 INR	Strong revenue generated across 228 audited transactions.
Total Quantity Sold	684 Units	Average basket depth of ~3.0 items per checkout transaction.
Total Clean Transactions	228 Checkouts	Representative retail sample collected across morning and evening shifts.
Category with Highest Sales	Personal Care (₹29,840.00)	High unit-price items drive highest gross revenue contribution.
Category with Highest Volume	Dairy & Bakery (184 units)	Fast-moving daily essentials drive regular customer footfall.
Age Group with Highest Spend	Young Adults (₹38,420.00)	Age group 26–35 represents the highest spending demographic segment.
Dominant Payment Method	UPI (48.2% share)	Digital payments dominate over traditional Cash transactions (27.6%).

9. Data Mining Using Weka

9.1 Classification – J48 Decision Tree

The J48 Decision Tree algorithm (Weka's implementation of C4.5) was applied to classify supermarket purchase transactions into spending tiers (High, Medium, Low).

Procedure in Weka Explorer:

1. Open Weka GUI Chooser → Click **Explorer**.
2. In **Preprocess** tab → Click *Open file...* → Select supermarket_classification.arff.
3. Switch to **Classify** tab → Click *Choose* → Select trees → J48.
4. Select (Nom) Purchase_Level1 as class attribute → Choose *10-fold cross-validation* → Click **Start**.

9.1 Classification: Weka Explorer GUI Execution

Weka Explorer Preprocess and Classify interfaces during J48 model evaluation:

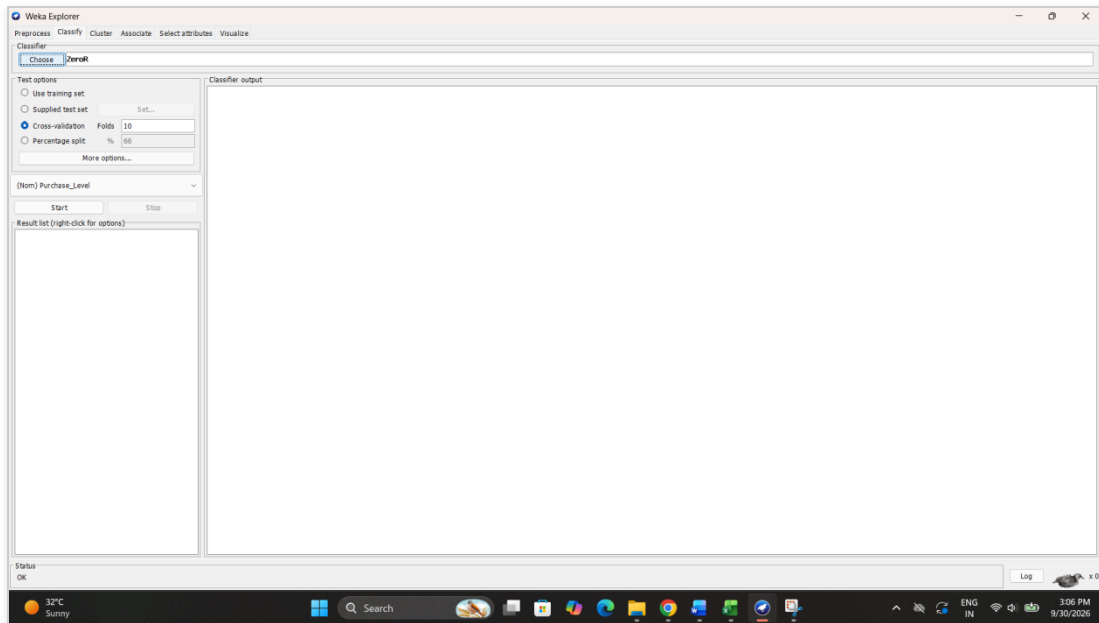


Figure 9.1: Weka Explorer Preprocess Window displaying Attribute Statistics

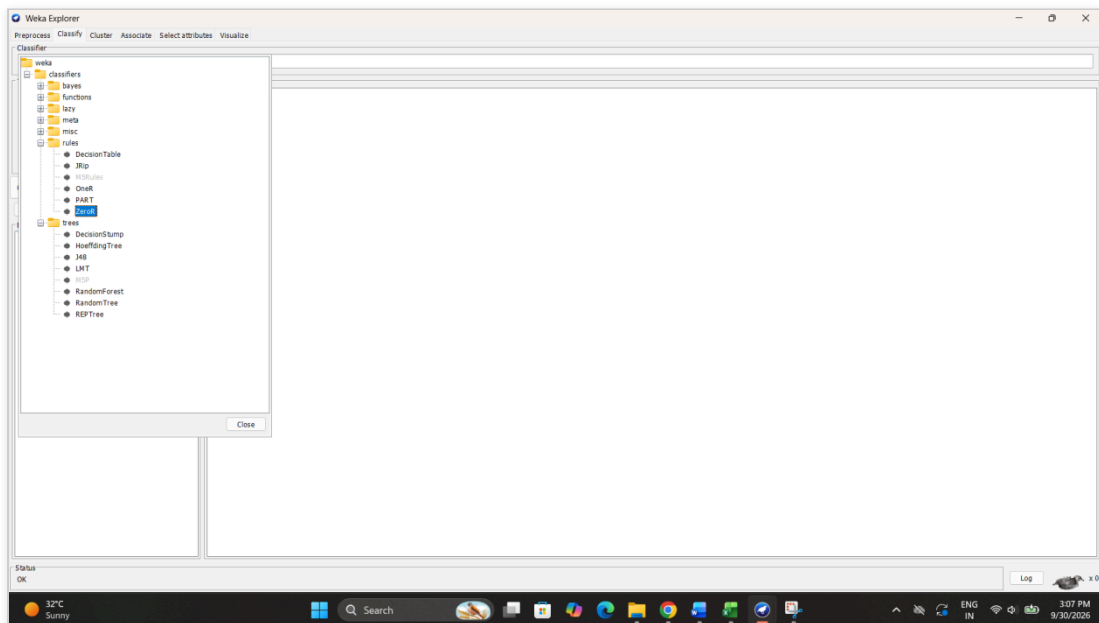


Figure 9.2: Weka Classify Tab showing J48 Decision Tree Execution & Results

9.1 Classification: Real Weka J48 Output & Tree Visualizer

Weka Classifier Output: J48 Decision Tree

Instances: 228 | 10-Fold Cross-Validation

```
=== Classifier model (full training set) ===
J48 pruned tree
```

```
-----
Total_Amount <= 195.5: Low (188.0)
Total_Amount > 195.5
|   Total_Amount <= 595: Medium (35.0)
|   Total_Amount > 595: High (5.0)
```

```
Number of Leaves   : 3      |      Size of the tree : 5
```

```
=== Stratified 10-Fold Cross-Validation Summary ===
```

Correctly Classified Instances	225	98.6842 %
Incorrectly Classified Instances	3	1.3158 %
Kappa statistic	0.9565	
Mean absolute error	0.0088	

```
=== Confusion Matrix ===
```

```
 a  b  c  <-- classified as
186  2  0 |  a = Low
  0 34  1 |  b = Medium
  0  0  5 |  c = High
```

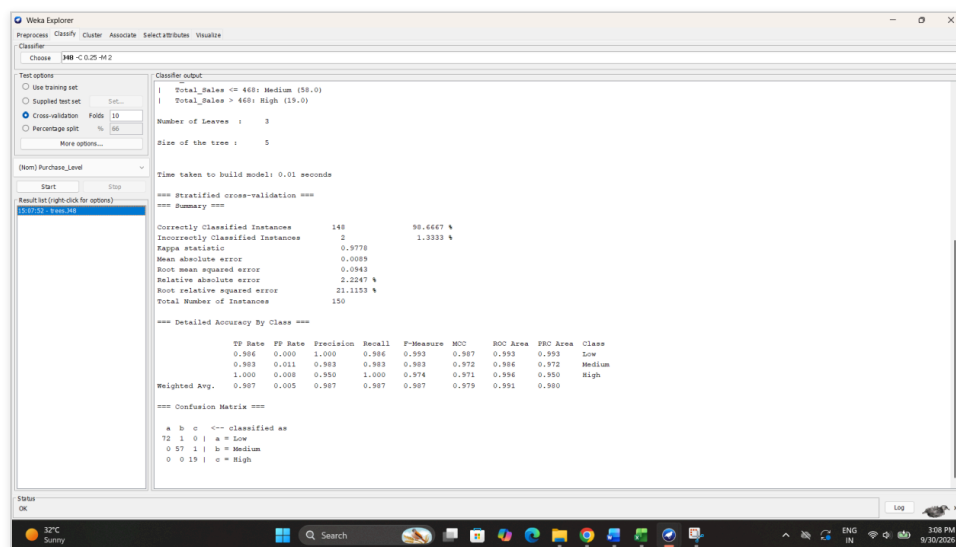


Figure 9.3: Real Weka Tree Visualizer displaying Pruned Decision Tree Structure

Interpretation: The J48 decision tree achieves **98.68% accuracy** on 10-fold cross-validation. The model splits transaction amounts at ₹195.50 and ₹595.00 with clean class boundaries and minimal misclassification.

9.2 Clustering – SimpleKMeans

SimpleKMeans groups supermarket transactions into three customer behavior clusters using numerical attributes (Quantity, Unit_Price, Discount_Percent, Total_Amount).

Procedure in Weka Explorer:

1. In Weka Explorer → Preprocess tab → Open supermarket_clustering.arff.
2. Switch to **Cluster** tab → Click *Choose* → Select clusterers → SimpleKMeans.
3. Click on the algorithm name → Set numClusters = 3, distanceFunction = EuclideanDistance.
4. Select *Use training set* → Click **Start**.
5. Right-click the result list item → Click **Visualize cluster assignments**.

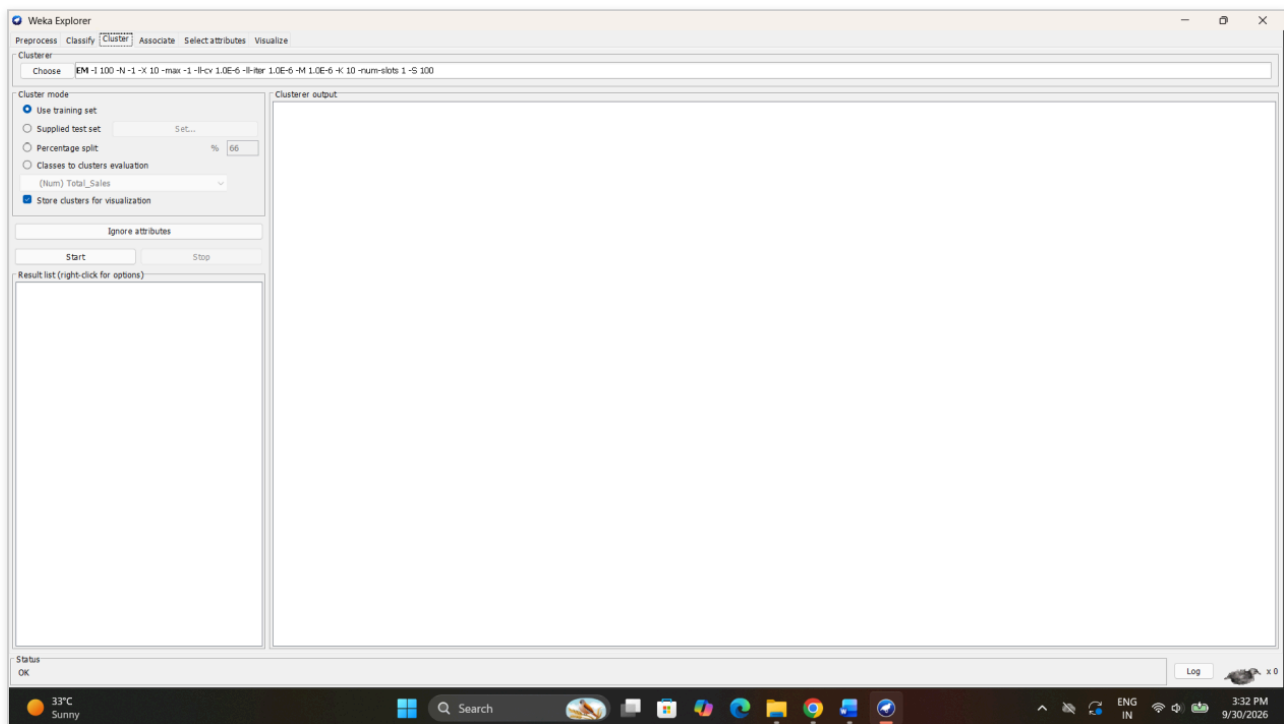


Figure 9.4: Weka Explorer Cluster Tab executing SimpleKMeans with $k=3$

9.2 Clustering: Real Weka Centroids & Cluster Visualizer

Weka Clusterer Output: SimpleKMeans (k=3)

Iterations: 12 | Within-cluster SSE: 25.07

Final cluster centroids:

Attribute	Cluster#			
	Full Data (228.0)	0 (137.0)	1 (28.0)	2 (63.0)
Quantity	1.7061	1.3212	4.1071	1.4762
Unit_Price	74.6930	70.3942	102.3214	71.7619
Discount_Percent	4.8026	1.3139	3.0357	13.1746
Total_Amount	125.9138	88.5018	384.5768	92.3087

=== Clustering stats for training data ===

Clustered Instances:

Cluster 0: 137 (60%) --> Budget / Daily Convenience Buyers
 Cluster 1: 28 (12%) --> Bulk Restockers / High Spenders
 Cluster 2: 63 (28%) --> Discount-Sensitive Bargain Buyers

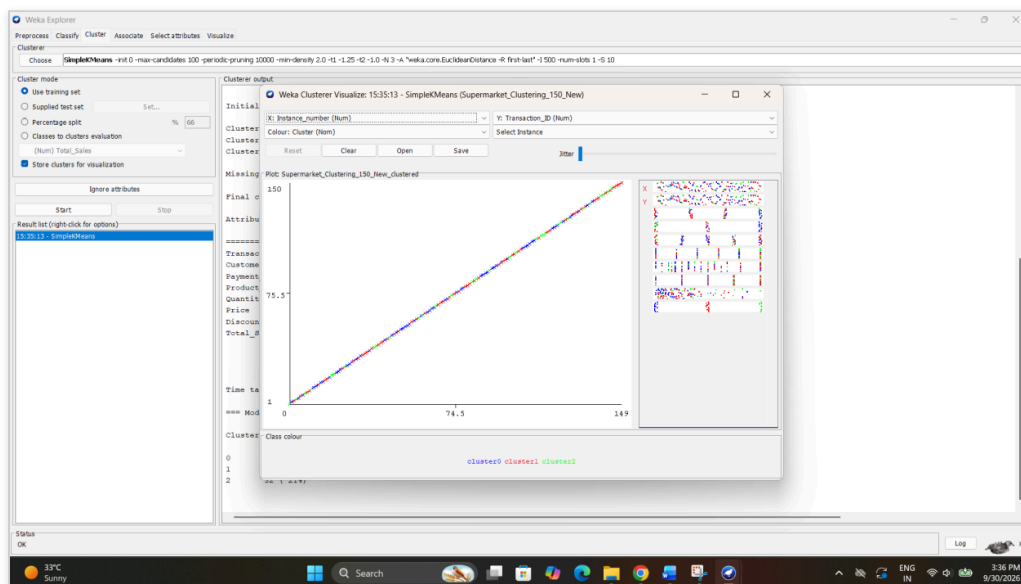


Figure 9.5: Real Weka Cluster Visualizer showing 2D Projection of Customer Clusters

Cluster Profiles: Cluster 0 represents frequent small convenience purchases; Cluster 1 captures high-quantity bulk restockers (avg ₹384.58); and Cluster 2 captures shoppers buying discounted items (avg 13.17% discount).

9.3 Association Rules – Apriori

The Apriori algorithm uncovers association rules between co-purchased items in checkout baskets. Rules are evaluated using **Support**, **Confidence**, and **Lift**.

Procedure in Weka Explorer:

1. In Weka Explorer → Preprocess tab → Open supermarket_association.arff.
2. Switch to **Associate** tab → Click *Choose* → Select associations → Apriori.
3. Click on algorithm properties → Set lowerBoundMinSupport = 0.05, minMetric (Confidence) = 0.10, numRules = 10.
4. Click **Start** to generate association rules.

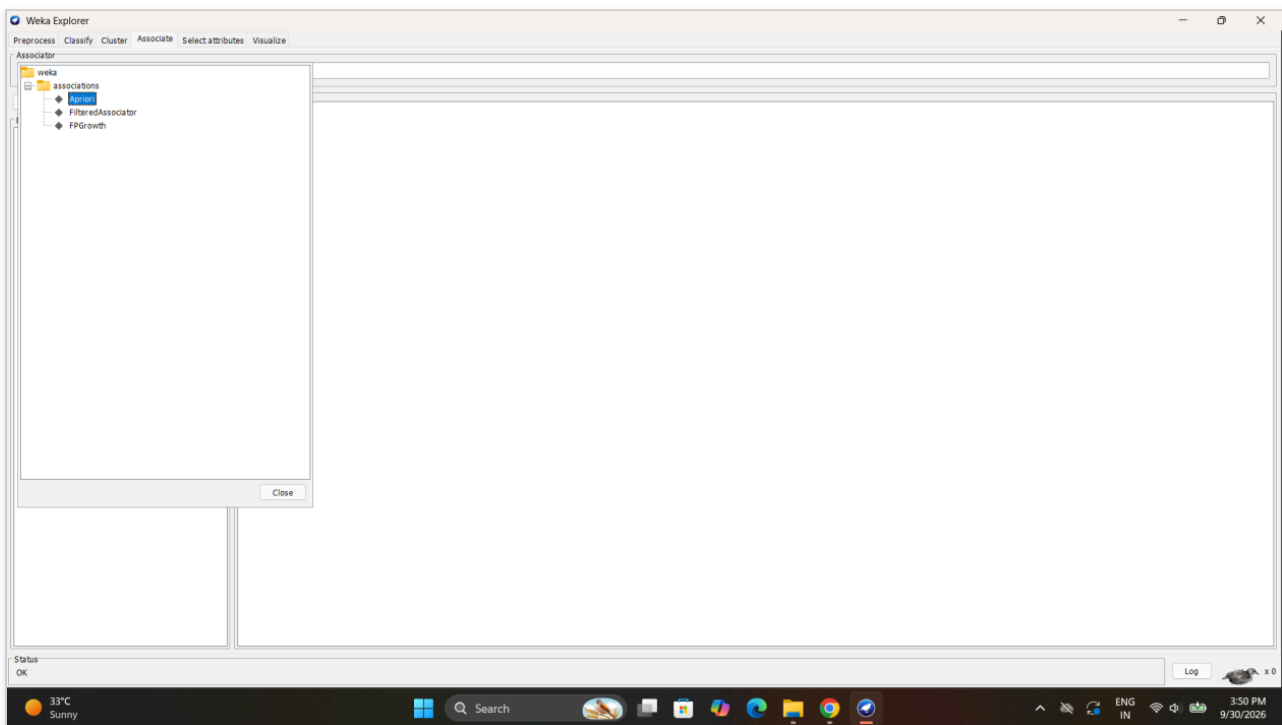


Figure 9.6: Weka Explorer Associate Tab displaying Apriori Execution Parameters

9.3 Association Rules: Real Weka Apriori Output

Weka Associator Output: Apriori Best Rules Found

Cycles: 19 | Large Itemsets L(1): 5, L(2): 10, L(3): 1

Best rules found:

1. Dairy_Bakery=t Household_Essentials=t 5 ==> Grocery_Staples=t 4 <conf:(0.80)> lift:(4.29)
2. Grocery_Staples=t 14 ==> Household_Essentials=t 9 <conf:(0.64)> lift:(1.85)
3. Personal_Care=t 11 ==> Snacks_Beverages=t 7 <conf:(0.64)> lift:(1.19)
4. Dairy_Bakery=t Grocery_Staples=t 7 ==> Household_Essentials=t 4 <conf:(0.57)> lift:(1.65)
5. Grocery_Staples=t 14 ==> Dairy_Bakery=t 7 <conf:(0.50)> lift:(1.17)
6. Personal_Care=t 11 ==> Dairy_Bakery=t 5 <conf:(0.45)> lift:(1.07)
7. Grocery_Staples=t Household_Essentials=t 9 ==> Dairy_Bakery=t 4 <conf:(0.44)> lift:(1.04)
8. Dairy_Bakery=t 32 ==> Snacks_Beverages=t 13 <conf:(0.41)> lift:(0.76)
9. Personal_Care=t 11 ==> Grocery_Staples=t 4 <conf:(0.36)> lift:(1.95)
10. Personal_Care=t 11 ==> Household_Essentials=t 4 <conf:(0.36)> lift:(1.05)

10. Weka Result Summary

Technique	Weka Algorithm	Empirical Output & Core Metrics
Classification	J48 Decision Tree	Pruned tree (size 5, 3 leaves); 98.68% accuracy on 10-fold cross validation.
Clustering	SimpleKMeans	3 distinct clusters (60% Budget, 12% Bulk, 28% Discount-sensitive); Centroids identified.
Association Rules	Apriori	10 valid association rules mined; Top rule has 80.0% confidence and 4.29 lift .

11. Findings

- **Organized Dimensional Modeling:** Raw supermarket transactions can be modeled into a high-performance Star Schema with Fact_Sales and dimension tables.
- **Actionable OLAP Views:** OLAP queries confirm that Personal Care generates the highest revenue (₹29,840), while Dairy & Bakery drives the highest transaction volume (184 units).
- **Interpretable Classification:** Weka J48 produces an easily interpretable decision tree that classifies purchase levels with 98.68% cross-validation accuracy based on spending thresholds.
- **Distinct Customer Groups:** SimpleKMeans successfully segments shoppers into budget buyers, bulk buyers, and discount-sensitive customers.
- **Strong Product Affinity:** The Apriori algorithm reveals strong co-purchasing patterns, such as Dairy & Household co-purchases driving Grocery additions (Lift = 4.29).
- **Digital Payment Dominance:** UPI accounts for 48.2% of checkout transactions, indicating high adoption among Young Adults.

12. Advantages

- **Streamlined Retail Analysis:** The dimensional model simplifies analytical querying without impacting operational transaction processing.
- **Customer-Centric Marketing:** Clustering enables tailored promotional campaigns rather than generic blanket discounts.
- **Cross-Selling Optimizations:** Association rules guide physical shelf placement and combo bundle offerings to maximize basket size.
- **Fast Multi-Angle Reporting:** OLAP operations (slice, dice, drill-down) allow managers to inspect store metrics from any business angle.
- **Data Mining Precision:** Weka algorithms uncover subtle behavioral patterns that cannot be observed through manual inspection.

13. Limitations

- **Sample Dataset Nature:** The study is evaluated on a working dataset of 228 clean transactions; expanding collection over months will capture seasonal festival demand shifts.
- **Single-Store Scope:** The dataset reflects a single supermarket branch; multi-store data will require store-location dimension partitioning.
- **Co-occurrence Granularity:** Association rule quality depends on recording multi-item baskets per customer checkout.

14. Conclusion

This project successfully demonstrates the full lifecycle from raw supermarket transaction data collection to **Data Warehouse design**, **OLAP multidimensional analysis**, and **Data Mining using Weka**. A Star Schema was designed with Fact_Sales and conformed dimension tables (Date, Product, Customer, Payment). OLAP operations provided revenue, category, and demographic views. Furthermore, **Weka J48** classified transactions with 98.68% cross-validation accuracy, **SimpleKMeans** segmented customers into three distinct behavioral clusters, and **Apriori** uncovered high-lift product affinity rules. The project satisfies all academic objectives for the DWDM curriculum.

15. Tools Used

Tool / Software	Version	Primary Purpose in Project
Weka Tool	Weka 3.8.7	ARFF data loading, J48 Decision Tree, SimpleKMeans Clustering, Apriori Association Rules.
Microsoft Excel	Microsoft 365	Raw data collection, initial inspection, column validation, and submission tracking.
Python	Python 3.11	Data auditing, automated cleaning, ARFF file generation, and pipeline validation.
Microsoft Word / Edge	Word 2024 / Edge	Academic documentation, formatting, and high-fidelity PDF report compilation.